

Documented failures concentrate at the problem-definition layer

no documented fitting failure in 36 batches · revision concentrates where new abstractions are defined

Decision layers

1 · Geometric fitting

fit + validate run in every batch; no documented fitting failure

STRONG · 36/36

2 · Problem definition

comparison domain · reuse boundary · representation · task selection

REVISED IN PLACE · 5 episodes

3 · Cross-batch abstraction

B15 shared spool carried a wrong extent into later installations

PROPAGATION RISK

Batches with ≥2 revisions, by band

B01–B06 · local fits

4/6

67%

B07–B19 · repeated equipment

13/13

100%

B20–B30 · connected systems

5/11

45%

B31–B36 · site closure

5/6

83%

visible _rN tags · filename-based lower bound, read as a distribution